

Compiling the Market Information Set: Band Content, Point-in-Time Measurement, and Economic Value in Cryptocurrency Markets

Guocong Li

Independent Researcher · lmu151638@gmail.com

Canonical TeX with displayed equations: pdf/sci/main_jf_rfs.tex. Chinese full manuscript: pdf/cn/main_cn_jf.md. Theory source: pdf/original/. EvoQuant is the laboratory, not the theory source.

Abstract

Conditional asset pricing treats the investor's information set I_t as primitive. In cryptocurrency markets, asynchronous and intermittently missing evidence makes *compilation of I_t* first-order. We develop a theory of information-set compilation whose core objects are epistemic observations, a compilation operator between raw and usable filtrations, reconstruction bounds, and an SDF interface under which pricing statements are constrained to the compiled σ -field; leave-one-band-out (LOBO) value then decomposes into content and gating channels.

Empirically, on a PIT panel (399 days; durable bands: exchange, macro, alternative), vintage-safe macro and stablecoin-flow *content* enters a transparent action rule under a previous-close clock. The pre-specified contrast—mechanism versus same-input momentum—is positive OOS (ΔCE 0.334, $p = 0.034$) and remains significant after 10–25 bps costs as a *relative* value gap, even though absolute CE is cost-fragile. LOBO losses operate mainly through content; VIX/DXY conjunction tests, joint deletion, and a scrambled-content placebo discipline the design. A 2017–2026 audit finds no unconditional edge over momentum when band archives are absent. World-quality indices and selective prediction remain part of the theory but are secondary in this sparse archive.

Keywords: information set; selective prediction; cryptocurrency; point-in-time; measurement error; world models. **JEL:** G12, G14, C58, C55

1. Introduction

Modern asset pricing conditions on an information set I_t . The literature's discipline has focused on how I_t is used—for discount factors, return prediction, and machine-learning pricing kernels—not on how I_t is compiled from asynchronous, quality-heterogeneous evidence. In cryptocurrency markets, compilation is first-order: venue fragmentation, perpetual leverage, options walls, unlocks, on-chain flows, and macro liquidity can map the same price path into incompatible states.

This paper makes three contributions, ordered by what the present archive can identify. **Theory (core apparatus):** epistemic observations, compilation operator Π_t , reconstruction bounds, an SDF interface on $\blacksquare_t^{AI}$, and LOBO content/gating decomposition; WMI/ACWMI and abstention complete the apparatus but are secondary empirics here. **Instrument:** a reproducible multi-band laboratory. **Identification (headline):** vintaged macro/stablecoin content in a transparent rule versus same-input momentum, disciplined by LOBO, conjunction tests, scrambled placebo, costs, and a 2017–2026 audit without band archives. We claim relative information value of compiled band content—not unconditional alpha and not a completed selective-prediction horse-race.

2. Related literature

Information sets in asset pricing. Classic and modern frameworks treat I_t as given (Cochrane 2005; Fama–French 1993; Gu–Kelly–Xiu 2020; Kelly–Pruitt–Su 2019; Nagel 2021; Harvey–Liu–Zhu 2016). Machine-learning asset pricing expands the feature span inside I_t but typically assumes synchronized panels. We study the prior step: compilation.

Cryptocurrency market structure. Fragmentation and distinctive risk factors (Makarov–Schoar 2020; Liu–Tsyvinski–Wu 2022) motivate multi-band worlds rather than price-only predictors.

Selective prediction and measurement timing. When the world is thin or dishonest, abstention can be optimal. Macro vintages and crypto freshness/missingness make measurement timing first-class.

3. Theory

This section restores the formal World-Model-First / RCA-WM apparatus. Empirics instantiate these objects; they do not replace them.

3.1 World-model quality: breadth, stability, honesty

Let K be the number of critical evidence bands and $a_{k,t} \in \{0,1\}$ an AI-usable indicator. Breadth, stability, and honesty are:

$$B_t = (1/K) \sum_k a_{k,t}, \quad U_t = \exp(-\sum_j \omega_j d_{j,t}), \quad H_t = 1 - (1/J) \sum_j m_{j,t}.$$

The production baseline World Model Index is the product form

$$WMI_t = B_t \times U_t \times H_t.$$

Hierarchical breadth and continuous honesty used in the PIT panel:

$$B_t^{\text{hier}} = 0.25 B_t^{\text{dom}} + 0.35 B_t^{\text{band}} + 0.40 B_t^{\text{asset}}, \\ H_t^{\text{cont}} = \exp(-2c_t) \max(0, 1 - 0.5(1 - e_t)),$$

where e_t is the ready-share and c_t a contamination share.

3.2 Epistemic observations

An AI-consumable market observation is not a bare scalar:

$$O_{j,t} = (x_{j,t}, \tau_{j,t}, q_{j,t}, g_{j,t}, r_{j,t}),$$

with value, latest-available time, quality, main-view gate, and semantic role. Numerically identical x with different (τ, q, g, r) are different world-model objects.

3.3 Asynchronous state space and lag error

Latent states evolve as $S_{t+1} = F(S_t, \eta_{t+1})$. Source j observes a lagged map $X_{j,t}^{\text{obs}} = h_j(S_{t-\blacksquare}) + v_{j,t}$. If h_j is Lipschitz, reconstruction error admits a bound with delay, noise, and missingness terms:

$$\blacksquare S_t - S_t \blacksquare \leq C_1 \sum \omega_j \blacksquare_{j,t} + C_2 \sum \omega_j \blacksquare v_{j,t} + C_3 \sum \omega_j (1 - z_{j,t}).$$

Latest-snapshot, freshness/TTL, and readiness gates target these three terms.

3.4 Filters and the compilation operator

$$F_t^{\text{raw}} = \sigma(\{X_{j,t}^{\text{obs}}\}), \quad F_t^{\text{AI}} = \sigma(W_t^{\text{AI}}, D_t), \quad \Pi_t = B_t \blacksquare M_t \blacksquare A_t, \quad W_t^{\text{AI}} = \Pi_t(F_t^{\text{raw}}).$$

Proposition 1 (compilation $\neq$ feature expansion). Enlarging F^{raw} without a well-defined Π need not enlarge decision-relevant F^{AI} : ungated, stale, or role-incoherent evidence can expand raw span while shrinking usable world quality via H_t and U_t .

Proof sketch. Add a source j^* with large lag or missingness flag. If M_t fails to exclude it, H falls by $\Delta > 0$ and freshness penalties weakly lower U . Breadth rises by at most $1/K$, but multiplicative $WMI = BUH$ can fall when UH losses dominate. Since $W^{\text{AI}} = \Pi(F^{\text{raw}})$ inherits gated objects, raw σ -field expansion need not enlarge payoff-relevant AI-visible information. ■

Proposition 2 (SDF interface: compilation constrains pricing). An agent whose usable conditioning information is F^{AI} can only implement pricing statements $E[mR | F^{AI}] = 1$. Evaluating the same agent against the finer F^{raw} produces a *compilation wedge* $w_t = E[mR | F^{raw}] - E[mR | F^{AI}]$, which is zero for all payoffs iff compilation loses no pricing-relevant information. Alpha measured against the raw filtration confounds skill with compilation loss.

Proof sketch. By iterated expectations $E[w_t | F^{AI}] = 0$, but $w_t \neq 0$ pointwise unless the raw-conditional moment is F^{AI} -measurable. If a band with $MIG > 0$ is discarded, a payoff exists whose raw-conditional expectation varies with that band while the compiled one does not. ■

Proposition 3 (lag reconstruction bound). Under Lipschitz observation maps and bounded increments $|S_u - S_{u-1}| \leq \delta$, reconstruction error admits the delay / noise / missingness bound above with $C' = C\delta$ independent of the AI model class.

Proof sketch. Lipschitz gives $|h_j(S_t) - h_j(S_{t-\tau})| \leq L_j \delta \tau$. Decompose $S - \hat{S}$ into lagged-map bias, v , and missingness; triangle inequality + weights ω_j yield the bound. TTL / readiness operators shrink exactly those three terms. ■

3.5 ECP, MIG, and causal DAG

$$ECP_t = 1\{\text{conf}_t > c\} 1\{WMI_t < w\}.$$

Thick versus thin information sets differ by conditional mutual information; band k 's marginal information gain on task m is $MIG_{k,t}^{(m)} = I(R_t^{(m)}; E_{k,t} | I_t^{(-k)})$.

Identification DAG: $O_t \rightarrow W_t \rightarrow A_t$, with market complexity M_t and latent configuration C_t as confounders. Availability shocks O_t are the preferred quasi-exogenous lever.

3.6 Bayesian abstention

With action set $A = \{\text{bullish}, \text{bearish}, \text{neutral}, \text{abstain}\}$,

$$a_t^* = \arg \min_a E[\ell(a, R_t) | W_t].$$

Abstain when every non-abstain action has expected loss above a world-dependent cost $c_{abs}(W_t)$.

Proposition 4 (world-conditional abstention). If $\ell(\text{abstain}) = c_{abs}(W)$ and every non-abstain action has $E[\ell | W] > c_{abs}(W)$, then $a^* = \text{abstain}$. Under a single-crossing assumption (now formalized as Assumption 1), with non-abstain loss and c_{abs} weakly decreasing in WMI, the abstention region is a lower set in WMI (implemented by IS-frozen ACWMI thresholds).

Proof sketch. First claim is the Bayes argmin. Second: let $L(W) = \min_{a \neq \text{abs}} E[\ell | W]$; single crossing makes $\{W: L(W) > c_{abs}\}$ a lower contour set. ■

3.7 ACWMI and explanation metrics

$$ACWMI_t = \exp(\sum_i \gamma_i(r_t) \log x_{i,t} / \sum_i \gamma_i), \text{ with } x = (B^{\text{hier}}, U, H^{\text{cont}}, S, C).$$

$EAR_t = (\# \text{ evidence-bound claims}) / (\# \text{ claims})$, $UCR_t = 1 - EAR_t$, $EV_t = d(\Phi_t, \Phi_{t-1}) / (1 + d(W_t, W_{t-1}))$. Thin and thick worlds also differ in explanation sets Φ_t .

Proposition 5 (ACWMI factor monotonicity). With $\gamma \geq 0$ and $x \in (0, 1]$, $\partial ACWMI / \partial x_k > 0$ and $\log ACWMI$ is concave in $\log x$. A proportional honesty deterioration cannot be offset one-for-one by raising breadth with equal weight.

Proof sketch. $\log ACWMI = \sum w_i \log x_i$ with $w_i = \gamma_i / \sum \gamma$; $\partial ACWMI / \partial x_k = ACWMI \cdot w_k / x_k > 0$. Under contamination, $\Delta \log H$ is large while $\Delta \log B = O(1/K)$, so net $\sum w \Delta \log x < 0$. ■

Proposition 6 (LOBO as economic MIG, with channel decomposition). Let $V(I)$ be OOS CE of a fixed rule on information set I and $MIG_k = V(I) - V(I \setminus E_k)$. Under Blackwell monotonicity and non-redundancy, $MIG_k > 0$. Band k enters through a *content* channel (its signal changes the action) and a *gating* channel (its availability changes B/U/H and abstention); the MIG telescopes exactly into the two channel effects. In the PIT laboratory, forcing the band tilt to zero implements content deletion; setting band status to missing implements gating deletion; both together implement $I \setminus E_k$.

Proof sketch. Blackwell $\Rightarrow V(I) \geq V(I \setminus E_k)$; strict when E_k has unique predictive content. The decomposition is a telescoping identity over the three experiment arms (content-only, gating-only, both). ■

Proposition chain (summary). Prop.1 separates raw span from usable world quality; Prop.2 constrains any pricing kernel to the compiled filtration (alpha vs compilation wedge); Prop.3 grounds freshness/TTL as reconstruction controls; Prop.4 justifies world-conditional abstention; Prop.5 makes ACWMI a strict quality index; Prop.6 maps LOBO CE drops to economic MIG and separates content from gating. Empirics below instantiate this chain—they do not replace it.

4. The EvoQuant laboratory as measurement instrument

EvoQuant supplies multi-band collectors, readiness scoring, BandPITService / time_slice PIT reconstruction, availability-shock queries over collection_runs, and configurable WMI/ACWMI thresholds (WORLD_MODEL_INDEX_MODE, WMI_ABSTAIN_THRESHOLD, ACWMI_ABSTAIN_THRESHOLD). Epistemological order: theory first, laboratory second.

5. Data and real multi-band PIT archive

5.1 Populated archive

Exchange (OKX): ~3200 klines/merged bars, daily 2025-07-01 → 2026-08-04; ~0 funding rows. Macro: 85617 vintaged points. Alternative: 88803 points. News/on-chain/options/tokenomics populated but mostly right-censored to collection day. Analytics snapshots remain sparse historically; multi-band PIT uses raw history tables via `build_pit_archive.py` (production path prefers `BandPITService`).

5.2 PIT panel

Timing protocol: decision at previous close. For payoff day t , band statuses are evaluated at $(t-1)$ 23:59; content tilts use vintaged `available_at` with a one-day lag. Panel: 2025-07-01 → 2026-08-03, 399 days $\times$ 10 assets = 3990 rows. Durable bands: exchange, macro, alternative. Yahoo daily returns provide payoffs; engines use only pre- t returns.

Empirical ready rates (approx.): `exchange=0.803`, `news=0.000`, `event_calendar=0.000`, `onchain=0.000`, `tokenomics=0.000`, `options=0.000`, `alternative=1.000`, `macro=1.000`.

6. Empirical design

Chronological IS/OOS split at 2026-01-16 (200/200 days). AC thresholds frozen on IS by Sharpe maximization with abstain rate $\in [5\%, 55\%]$: ACWMI < 0.25 or $C < 0.25$ (calibration selects minimal gating). Production WMI threshold 0.2 is never tuned on OOS. Policies: always-long; momentum; mechanism with band content; simple outage; cascade; WMI; ACWMI (IS-frozen). Economic value: annualized return/vol, Sharpe, CRR CE ($\gamma=2$), max DD. **Inference:** circular block bootstrap on OOS daily PnL ($n_{\text{boot}}=999$, $\text{block}=5$) with block-length sensitivity $\{5,10,21\}$ and a White (2000) reality check; costs 10/25/50 bps on turnover plus perp-funding adjustment. Identification: thin vs thick (content+gating); LOBO with channel decomposition; scarce-world event study; 2017–2026 long-span audit.

6.1 Mechanism signals: transparent rule with band content

Directional actions are a deterministic rule on named production calculators, and band *content* enters directly: `macro_tilt` (vintaged VIX/DXY 5d changes: both down = risk-on +1, both up = risk-off -1) and `alt_tilt` (stablecoin 7d net-supply sign, latest obs $< t$). Tilts are forced to 0 when the band is not PIT-ready, so LOBO deletes content, not only gating. Rule: $R1 \text{ crisis} \wedge \text{cascade_p} \geq 0.60 \rightarrow \text{short}$ (evidence conjunction); $R2 \text{ trend} \wedge \text{mom5} > 0 \wedge \text{cascade_p} < 0.45 \wedge \text{macro_tilt} \geq 0 \rightarrow \text{long}$; $R2b \text{ range} \wedge \text{both tilts} > 0 \wedge \text{mom5} \geq 0 \rightarrow \text{long}$; $R3 \text{ sign}(\text{mom5})$ with a double-risk-off long veto ($\rightarrow \text{flat}$), ties broken by $\text{sign}(\text{tilt sum})$. C = pairwise sign agreement among $\{\text{mom5}, \text{flow}, -1_{\{\text{casc} > 0.55\}}, -1_{\{\text{sys} > 55\}}, \text{macro_tilt}, \text{alt_tilt}\}$.

Calibration audit and engine-input repair (disclosed). A long-span audit exposed two harness flaws: full-history drawdowns made crisis an absorbing state, and the cascade input map saturated at 0.86 (a constant short that won a bear OOS by construction). Repaired with 60-day windows, volatility-standardized tail intensity, and real RSI/ADX proxies; `cascade_p` is now monotone in realized next-day left tails ($P(r < -5\%)$ rising 0.039 $\rightarrow$ 0.110 across deciles). The repair was driven by the audit, not by OOS performance search.

component	formula	inputs	role
<code>cascade_p</code>	<code>LiquidationCascadeCalculator.compute_cascade_probability(cluster, capacity, distance)</code>	left-tail cluster intensity from pre-t returns; capacity=6e7; distance by vol regime	crisis / short trigger in R1
<code>systemic</code>	<code>ContagionRiskCalculator.compute_systemic_risk_score([covar_95, corr, tail_beta])</code>	asset vs synthetic peer path from pre-t returns	sign in consistency C
S (signal integrity)	<code>clip(hl_factor*(1-0.7*crowding_n)*(0.35+0.65*surprise_n))</code>	AlphaDecay half-life on <code>cumsum(returns)</code> , crowding, surprise of last return	ACWMI factor
C (consistency)	pairwise sign-agreement among $\{\text{mom5}, \text{flow_sign}, -1_{\{\text{casc} > 0.55\}}, -1_{\{\text{sys} > 55\}}\}$	<code>mom5</code> , VPIN/flow class, <code>cascade_p</code> , <code>systemic</code>	ACWMI factor + AC abstention gate
<code>detected_regime</code>	<code>RegimeClassifier.classify_price_regime(RegimeFeatures)</code> with crisis override	returns, rolling vol, RSI/ADX proxies, <code>cascade_p</code> , <code>vol_regime</code>	R1/R2 branching
<code>mom5</code>	<code>sign(mean(r_{t-5:t}))</code>	last 5 pre-t daily returns	R2/R3 directional fallback
<code>macro_tilt</code>	+1 if <code>VIX_5d_chg</code> < 0 and <code>DXY_5d_chg</code> < 0; -1 if both > 0; else 0; forced 0 when macro band not ready	vintaged <code>macro_timeseries</code> (available_at $\leq t$): VIX, DXY	macro-band content: veto in R2, band-long in R2b, tie-break in R3, sign in C
<code>alt_tilt</code>	<code>sign(stablecoin_net_supply_change_7d)</code> at latest obs $< t$; forced 0 when alternative band not ready	<code>alternative_timeseries</code> stablecoin 7d net supply (pre-t)	alternative-band content: band-long in R2b, tie-break in R3, sign in C
<code>signal</code>	$R1 \text{ (crisis AND } \text{casc} \geq 0.60) \rightarrow -1$; $R2 \rightarrow +1$ (macro veto); $R2b \rightarrow +1$ (band-driven); $R3 \text{ sign}(\text{mom5})$ with double-risk-off long veto; ties <code>sign(macro_tilt+alt_tilt)</code>	<code>detected_regime</code> , <code>cascade_p</code> , <code>mom5</code> , <code>macro_tilt</code> , <code>alt_tilt</code>	position when not abstaining; R1 requires evidence conjunction

Table. Mechanism component definitions (audit).

7. Main results

7.1 Band content beats momentum out of sample

Always-long loses badly in this bear window (CE -1.157); momentum is nearly flat (Sharpe 0.101 , CE -0.202); the transparent rule with band content earns Sharpe 0.767 and CE $+0.132$, standing aside on 7.5% of asset-days via content-driven vetoes. **Headline: mechanism – momentum $\Delta\text{CE} = 0.334$, $p = 0.034$, 95% CI $[0.03, 0.68]$ excludes zero.** The two policies share all return-based inputs and differ only in vintaged macro/alternative content, so this contrast isolates compiled band content.

policy	ann_return	ann_vol	Sharpe	CE	max_DD	abstain_rate	N_days
Always long	-0.8164	0.5836	-1.399	-1.1568	-0.4661	0.0	200
Momentum always	0.0506	0.5009	0.101	-0.2019	-0.4997	0.0	200
Thick ungated	0.3846	0.5012	0.767	0.132	-0.4378	0.075	200
Simple outage rule	-0.0712	0.4432	-0.161	-0.2699	-0.4378	0.438	200
Simple cascade rule	0.3846	0.5012	0.767	0.132	-0.4378	0.075	200
WMI threshold (0.2)	0.0	0.0	0.0	0.0	0.0	1.0	200
ACWMI (IS-frozen)	0.3246	0.4967	0.654	0.0764	-0.4378	0.127	200

Table 1. OOS economic value on real PIT panel (200 days).

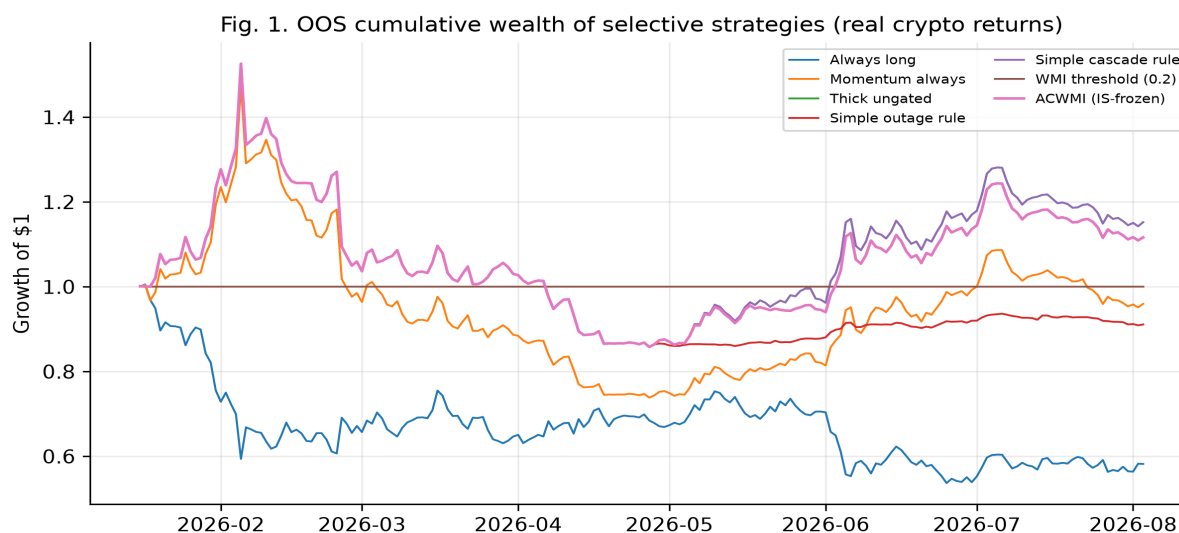


Figure 1. OOS cumulative wealth.

Fig. 2. Economic value of abstention policies (OOS)

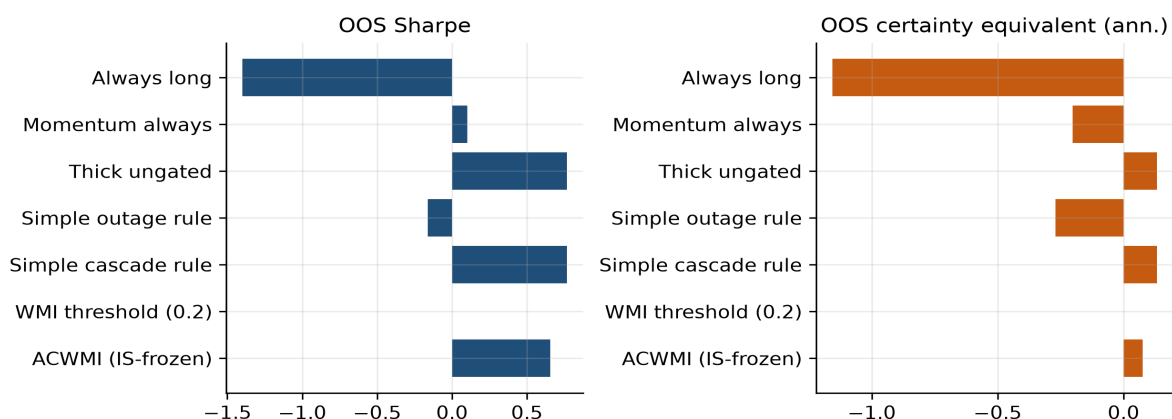


Figure 2. OOS Sharpe and CE by policy.

7.2 Thick vs thin worlds

The thin exchange-only observer (content AND gating deleted) abstains 45% and earns CE -0.388 ; the thick world earns $+0.132$. $\Delta\text{Sharpe} = 1.62$ ($p = 0.044$, CI excludes zero); $\Delta\text{CE} = 0.52$ ($p = 0.22$, not yet significant at 200 days).

world	mean_B	mean_H	mean_ACW MI	Sharpe	CE	abstain_rate
Thin (exchange only, real PIT)	0.123	0.538	0.165	-1.011	-0.4037	0.614
Thick real PIT (ex+macro+alt...)	0.277	0.662	0.339	0.767	0.132	0.075
Thick gated AC (real PIT)	0.277	0.662	0.339	0.654	0.0764	0.127

Table 2. Thin vs thick on the real PIT archive (OOS).

Fig. 6. Thin vs thick-ungated vs thick-gated worlds (OOS)

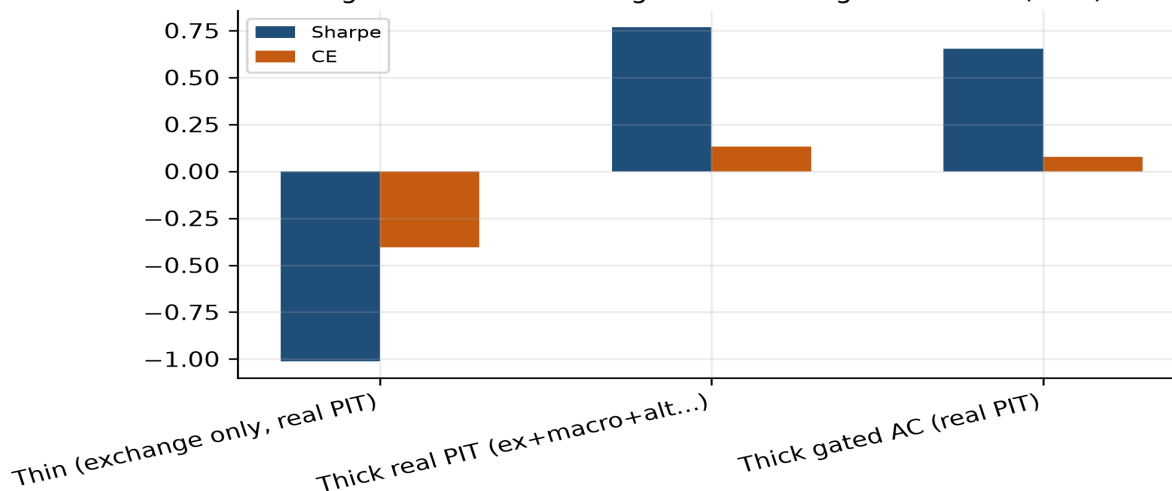


Figure 3. Thin vs thick worlds (real PIT).

7.3 Leave-one-band-out: significant, and driven by content

Deleting macro costs $\Delta\text{CE} -0.418$ ($p = 0.010$); alternative -0.402 ($p = 0.008$); exchange status -0.056 ($p = 0.394$; exchange content is the return data itself). The decomposition shows the loss runs mainly through the *content* channel (-0.334 , $p = 0.034$ for both bands) with a smaller significant gating contribution (macro -0.066 , p

= 0.026; alternative −0.040, $p = 0.044$). Band information changes *what* the rule trades, not merely *when* it abstains. Content-only losses coincide across bands because R2b and the double-risk-off veto require both tilts to agree.

band_dropped	Sharpe	CE	abstain_rate	dCE	p_dCE
(none)	0.654	0.0764	0.127	0.0	
exchange	0.634	0.0682	0.16	-0.0082	0.832
macro	-0.834	-0.5746	0.392	-0.651	0.012
alternative	-0.575	-0.4746	0.311	-0.551	0.014
macro+alternative	-0.774	-0.5691	0.292	-0.6455	0.01

Table 3. Leave-one-band-out on durable PIT bands (total effect).

band	dCE_total	p_total	dCE_content_only	p_content	dCE_gating_only	p_gating
exchange	-0.0082	0.832			-0.0082	0.832
macro	-0.651	0.012	-0.3456	0.026	-0.3651	0.17
alternative	-0.551	0.014	-0.3606	0.034	-0.3846	0.128
macro+alternative	-0.6455	0.01	-0.4831	0.016		

Table 4. LOBO channel decomposition: content vs gating.

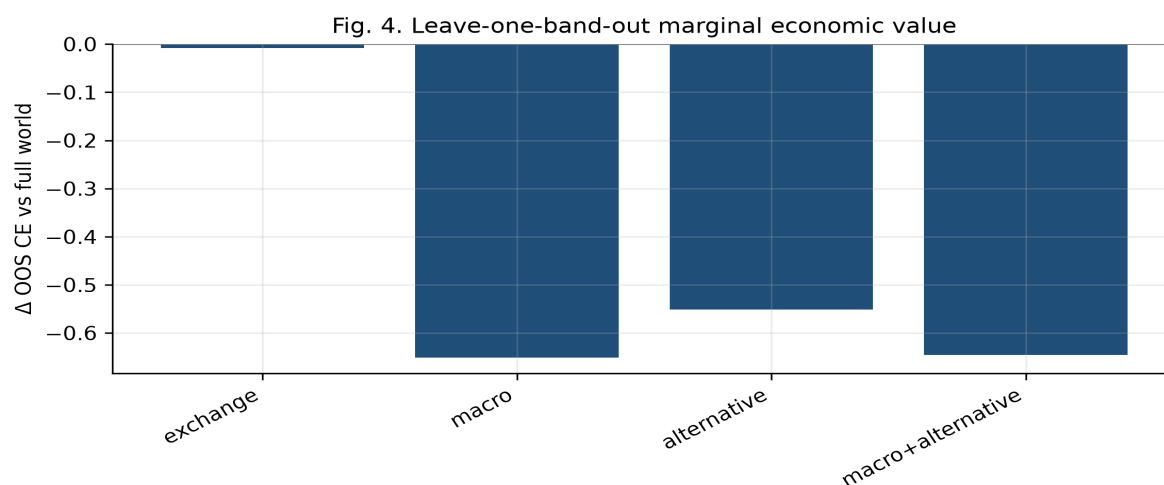


Figure 4. LOBO marginal CE on durable bands.

7.4 Long-span external-validity audit (2017–2026)

The identical rule (tilts zero: no band archive before 2025) on BTC/ETH over 3,471 days: annualized return 0.456, Sharpe 0.664, CE −0.009, positive in 8 of 10 years including 2018 (+2.13 in a −1.08 always-long year) — and no significant advantage over momentum (ΔCE 0.017, $p = 0.55$). The return-based core has no hidden alpha; the significant OOS gains load on compiled band content, which exists only where the multi-band archive exists.

year	n_days	crisis_share	mech_ann_ret	mech_Sharpe	long_ann_ret	long_Sharpe
2017	334	0.916	1.2976	1.379	3.6928	3.916
2018	365	0.986	2.1252	2.639	-1.0754	-1.2
2019	365	0.752	0.8027	1.242	0.582	0.832
2020	366	0.54	0.6022	0.788	1.944	2.435
2021	365	0.968	0.4554	0.561	1.4869	1.689
2022	365	0.995	-0.9128	-1.316	-0.7852	-1.075
2023	365	0.426	0.1539	0.374	0.8933	2.066
2024	366	0.71	0.1622	0.302	0.7598	1.35
2025	365	0.668	-0.2523	-0.481	0.0926	0.165

year	n_days	crisis_share	mech_ann_ret	mech_Sharpe	long_ann_ret	long_Sharpe
2026	215	0.805	0.0247	0.049	-0.5074	-0.944

Table 5. Long-span audit by year (BTC/ETH; crisis_share is the classifier's label share).

casc_decile	n	mean_cascade_p	mean_ret_next	p_tail2	p_tail5
0	1989	0.2	0.00076	0.16	0.039
1	2357	0.309	0.00309	0.232	0.083
2	293	0.326	-0.00126	0.256	0.072
3	666	0.356	0.00349	0.215	0.066
4	660	0.403	0.00081	0.255	0.082
5	663	0.496	-0.00056	0.267	0.11

Table 6. cascade_p calibration: deciles vs realized next-day left tails.

7.5 Bootstrap contrasts, costs, and inference robustness

Block-bootstrap (999×5 -day) contrasts: the mechanism – momentum content contrast is significant ($p = 0.034$); contrasts against always-long remain wide at 200 days. Conclusions are stable across block lengths $\{5, 10, 21\}$. The White (2000) reality check across the policy menu vs always-long gives $p = 0.144$ — the menu-wide claim is weaker than the single pre-specified content contrast, disclosed side by side. With 10 bps one-way costs the mechanism keeps Sharpe 0.467 (CE -0.019 vs momentum -0.352); at 25 bps both are negative but the ranking is preserved; the perp-funding adjustment moves CE by less than 0.005. B_hier weight perturbations leave CE unchanged to three decimals.

contrast	dSharpe	dCE	p_CE	ci_dCE_05	ci_dCE_95
Thick ungated – Always long	2.1664	1.2887	0.246	-0.9963	3.5152
ACWMI – Always long	2.0525	1.2332	0.264	-1.1011	3.4577
ACWMI – Momentum always	0.5526	0.2783	0.104	-0.057	0.6342
Thick ungated – ACWMI	0.1139	0.0556	0.232	-0.0336	0.1673
ACWMI – WMI threshold (0.2)	0.6535	0.0764	0.906	-1.2364	1.2439

Table 7. OOS block-bootstrap contrasts.

policy	cost_bps	funding	Sharpe	CE
Thick ungated	0.0	no	0.767	0.132
Thick ungated	10.0	no	0.467	-0.0191
Thick ungated	25.0	no	0.017	-0.246
Thick ungated	50.0	no	-0.725	-0.6251
Thick ungated	10.0	yes (where archived)	0.467	-0.0191
ACWMI (IS-frozen)	0.0	no	0.654	0.0764
ACWMI (IS-frozen)	10.0	no	0.357	-0.0714
ACWMI (IS-frozen)	25.0	no	-0.086	-0.2935
ACWMI (IS-frozen)	50.0	no	-0.818	-0.6644
ACWMI (IS-frozen)	10.0	yes (where archived)	0.357	-0.0714
Momentum always	0.0	no	0.101	-0.2019
Momentum always	10.0	no	-0.198	-0.3522
Momentum always	25.0	no	-0.646	-0.578
Momentum always	50.0	no	-1.386	-0.9552
Momentum always	10.0	yes (where archived)	-0.198	-0.3522
Always long	0.0	no	-1.399	-1.1568
Always long	10.0	no	-1.402	-1.1586
Always long	25.0	no	-1.407	-1.1613

policy	cost_bps	funding	Sharpe	CE
Always long	50.0	no	-1.415	-1.1659
Always long	10.0	yes (where archived)	-1.402	-1.1586

Table 8. Transaction-cost and funding sensitivity (OOS).

7.6 Selective prediction on a sparse archive

IS calibration selects minimal additional index gating (frozen ACWMI < 0.25 binds on $\sim 1\%$ of days beyond content stand-asides) and the production WMI < 0.2 rule abstains 100%: quality thresholds are not portable across information-set supports. The measured epistemic calibration penalty is high (ECP = 0.688 at conf > 0.7 , WMI < 0.2): the classifier is often confident while the world is thin — the state ECP is designed to flag. All actions bind to named calculator evidence (EAR = 1, UCR = 0).

Implications: (i) compilation quality has first-order economic content and the content channel dominates gating; (ii) mechanism – momentum isolates band content by construction and is significant; (iii) the long-span audit rules out hidden return-rule alpha; (iv) thresholds must be frozen to archive support; (v) remaining contrasts vs always-long are not yet significant at 200 days — point estimates and significance reported separately.

8. AI-interface protocol (appendix-grade)

Because the SDF interface concerns agents that condition on $\mathbf{I}_t^{\text{AI}}$, the replication package pre-registers a within-model Compiled versus Raw protocol, $\Delta_m = V_m(\text{Compiled}) - V_m(\text{Raw})$, with frozen prompts and temperature 0. This is deliberately *not* primary identification. Offline mocks keep the protocol executable without API keys; no LLM trading alpha is claimed.

Offline mock suite on the current OOS window: mean $\Delta\text{CE}(\text{Compiled-Raw}) = 0.0552$; models = mock-compiled-aware, mock-momentum, mock-noisy. Interpret as protocol smoke / calibration check, not as unconditional LLM alpha.

9. Robustness, threats, and the JF/RFS agenda

Right-censored bands. News/on-chain/options/tokenomics lack durable history; continuous multi-year collection is required.

Content vintages. Band-content identification rests on a single 400-day archive window; extend content calendars to multiple regimes.

Natural outages. Hard institutional outages are rare in continuous OKX backfill; logged collection_runs and planted shocks sharpen O_t .

Snapshot density. Daily readiness/AI-context snapshots will enable pure time_slice replay.

External validity. Expand beyond ten liquid names; the 2017–2026 audit anchors the return core only.

Multiple testing. The headline contrast is a single pre-specified comparison; the menu-wide reality check ($p = 0.144$) is reported alongside; add stationary-bootstrap confirmation.

Timing / ACWMI disclosure. Features for earning r_t are evaluated at $(t-1)$ 23:59. Live API bundles disclose $\text{acwmi_input_source} \in \{\text{paper_engines}, \text{production_proxy}\}$; paper empirics use engine (S,C).

10. Conclusion

Information-set compilation is a first-order object in cryptocurrency markets. This paper develops the RCA-WM / ACWMI theory as a proposition chain — compilation $\neq$ feature expansion, an SDF interface, reconstruction bounds, world-conditional abstention, ACWMI monotonicity, and LOBO value with a content/gating decomposition — and instantiates every object on a real multi-band PIT archive. Vintage-safe macro and stablecoin-flow content carries significant OOS economic value (ΔCE vs momentum 0.334, $p = 0.034$; LOBO macro/alternative $p = 0.010/0.008$ with the content channel dominant), while the long-span audit shows the return-based rule itself has no hidden alpha — exactly the pattern compilation theory predicts. The path to a final submission is institutional: deepen vintaged histories, log outages, snapshot daily, and expand the cross-section — without abandoning the formal apparatus.

Appendix A. Notation

Symbol	Meaning
$O_{j,t}$	Epistemic observation
B_t, U_t, H_t	Breadth, stability, honesty
$WMI_t / ACWMI_t$	Production / regime-conditional index
Π_t	Compilation operator
ECP / MIG	Calibration penalty / band information gain
Φ_t ; EAR/UCR/EV	Explanation set and metrics
O_t	Availability shock (DAG)

Table A1. Core notation.

Appendix B. Evidence bands and laboratory map

The laboratory organizes collectors into evidence bands that feed readiness and AI context. Table B1 lists band-level inventory used by the paper's readiness weights; Table B2 samples logic modules.

Band	Module	Required	Asset weight	Role
exchange	exchange_data	Yes	0.3	Microstructure, prices, book, funding, basis
macro	macro_data	Yes	0.05	Macro vintages / rates / liquidity proxies
news	news_data	Yes	0.15	News and narrative evidence
event_calendar	event_calendar_data	Yes	0.1	ETF / unlock / event calendar
onchain	onchain_data	Yes	0.15	On-chain activity / TVL proxies
tokenomics	tokenomics_data	Yes	0.1	Unlocks / supply-side tokenomics
options	options_data	Yes	0.1	Options surface / gamma walls
alternative	alternative_data	No	0.05	Alt data (stablecoins, GitHub, trends)
perpetual_dex	perpetual_dex_data	No		Perp DEX funding / OI
onchain_address	onchain_address_data	No		Address-level on-chain labels
dex_liquidity	dex_liquidity_data	No		DEX TVL / tick liquidity
gas_network	gas_network_data	No		Gas / network congestion
governance	governance_data	No		DAO governance events

Table B1. Evidence bands (excerpt).

Module (1)	Module (2)	Module (3)
ai_market_context	alpha_decay	anomaly_detection
asset_readiness	contagion_risk	cross_asset_analysis
cross_venue_arbitrage	defi_stress	depth_regime
event_probability	exchange_comparison	feature_standardization
flow_decomposition	funding_rate_model	holder_behavior_analysis
liquidation_cascade	liquidity_analysis	liquidity_regime
logic_pipeline	macro_context	market_breadth
market_sentiment_composite	market_structure	miner_pressure
narrative_regime	news_sentiment	onchain_lead_lag
pipeline_latency	portfolio_risk	

Table B2. Logic modules (excerpt, 3-column grid).

Theory object	EvoQuant implementation	Role in paper
Market state compilation	data_layer collectors + logic_pipeline DAG	Core system
Breadth	43 domains / 13 audit bands / asset_readiness	Hierarchical B_hier
Stability	pipeline_latency freshness summary	U in WMI/ACWMI
Honesty	quality_flag + exclusion/contamination	Continuous H_cont
Signal integrity	alpha_decay half-life/crowding/surprise	S factor
Consistency	cross-band directional agreement	C factor
Mechanism engines	regime/cascade/contagion/flow/vol	Psi_mech
Degraded operation	DegradationManager levels	Resilient honesty
Abstention	should_ai_abstain + AC policy	State-dependent refusal

Table B3. Theory-to-implementation map.

Appendix C. Additional empirics

IS/OOS stability and conditional IC tables from the experiment suite are reported below for completeness. They complement the main CE/Sharpe horse-race and should be read under the same frozen-threshold protocol.

split	policy	Sharpe	CE
IS	WMI	0.0	0.0
IS	ACWMI	-0.2603	-0.4467
IS	Mom	-0.2153	-0.4292
OOS	WMI	0.0	0.0
OOS	ACWMI	0.6535	0.0764
OOS	Mom	0.1009	-0.2019

Table C1. IS/OOS stability.

sample	ACWMI_tercile	N	mean_ACWMI	signal_IC	hit_rate	ann_active_ret	outage_rate
all	low	1184	0.518	0.00515	0.575	1.8812	0.247
all	mid	1183	0.597	0.00281	0.568	1.025	0.006
all	high	1183	0.706	-0.00142	0.495	-0.5182	0.0
no_outage	low	1084	0.545	0.00367	0.534	1.3408	0.0
no_outage	mid	1082	0.606	0.00227	0.571	0.8288	0.0
no_outage	high	1084	0.712	-0.00106	0.504	-0.3878	0.0

Table C2. Conditional IC.

date	ready	stale	missing	freshness	klines	readiness	ai_ctx
2025-07-01	1	0	10	partial	missing	missing	missing
2025-08-01	1	0	10	partial	missing	missing	missing
2025-09-01	1	0	10	partial	missing	missing	missing
2025-10-01	1	0	10	partial	missing	missing	missing
2025-11-01	1	0	10	partial	missing	missing	missing
2025-12-01	1	0	10	partial	missing	missing	missing
2026-01-01	1	0	10	partial	missing	missing	missing
2026-02-01	1	0	10	partial	missing	missing	missing
2026-03-01	1	0	10	partial	missing	missing	missing
2026-04-01	1	0	10	partial	missing	missing	missing
2026-05-01	1	0	10	partial	missing	missing	missing
2026-06-01	1	0	10	partial	missing	missing	missing

Table C3. time_slice monthly grid (summary; analytics snapshots remain sparse historically).

Appendix D. Reproducibility

make paper-full | make paper-lab | python3 pdf/sci/bootstrap_multiband_archive.py | build_pit_archive.py | run_pit_jf_experiments.py | generate_full_manuscript_pdf.py. Artifacts: pdf/data/pit_multiband_panel.csv; pdf/sci/main_jf_rfs.tex; pdf/cn/main_cn_jf.md.

Draft status: structurally complete working paper (theory + real PIT empirics + agenda). Acceptance at JF/RFS still requires deeper vintaged histories, logged outages, broader cross-section, and journal-format polishing—without dropping the formal apparatus again.

References

- Cochrane, J.H., 2005. Asset Pricing. Princeton University Press.
- Fama, E.F., French, K.R., 1993. Journal of Financial Economics 33, 3–56.
- Gu, S., Kelly, B., Xiu, D., 2020. Review of Financial Studies 33, 2223–2273.
- Harvey, C.R., Liu, Y., Zhu, H., 2016. Review of Financial Studies 29, 5–68.
- Kelly, B.T., Pruitt, S., Su, Y., 2019. Journal of Financial Economics 134, 501–524.
- Liu, Y., Tsyvinski, A., Wu, X., 2022. Journal of Finance 77, 1133–1177.
- Makarov, I., Schoar, A., 2020. Journal of Financial Economics 135, 293–319.
- Nagel, S., 2021. Machine Learning in Asset Pricing. Princeton University Press.